

# Chapter 26

## Review for the Final Exam

Lecture 26 | Devore 9e: Ch. 2–6

The final is cumulative, with the weight on Lectures 18–24 (joint distributions, CLT, estimation). This chapter is the review class: a checklist, a formula sheet, and the worked problems.

**Remark 26.1** (Scope). Both the Fall 2025 and Spring 2026 finals stopped at maximum likelihood — eight problems running Bayes, a named continuous distribution, joint/conditional distributions, covariance and correlation, the CLT, unbiased estimators, method of moments, MLE. **Confidence intervals (Lecture 25) were not examined.** If you change that, change this chapter too.

### 26.1 Checklist

| Topic                  | You should be able to                                     |
|------------------------|-----------------------------------------------------------|
| Bayes                  | invert a conditional; recognize the base-rate trap        |
| Transformations        | get $f_Y$ from $f_X$ via the cdf                          |
| Named distributions    | recognize the scenario and quote pdf, mean, variance      |
| Joint distributions    | normalize, marginalize, condition; <i>draw the region</i> |
| Covariance/correlation | compute from a joint pmf/pdf; interpret the sign          |
| CLT                    | approximate $\bar{X}$ ; know when it applies              |
| Estimation             | check unbiasedness; method of moments; MLE                |

## 26.2 Master formula sheet

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B \mid A_i)}{\sum_j \mathbb{P}(A_j)\mathbb{P}(B \mid A_j)}$$

$$\mathbb{E}[h(X)] = \int h(x)f(x) \, dx,$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|,$$

$$f_X(x) = \int f(x, y) \, dy$$

$$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)},$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

$$\text{Var}\left(\sum a_i X_i\right) = \sum a_i^2 \sigma_i^2 \text{ (independent)}$$

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n},$$

$$\bar{X} \overset{\text{approx}}{\sim} \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\mathbb{E}[S^2] = \sigma^2,$$

$$\mathbb{E}[X^k] = \overline{X^k} \text{ (method of moments)}$$

$$\ell(\theta) = \sum_i \log f(x_i; \theta) \text{ (maximize for the MLE)}$$

and two more that hold in general:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X], \quad \text{Var}\left(\sum_i a_i X_i\right) = \sum_{i,j} a_i a_j \text{Cov}(X_i, X_j).$$

## 26.3 Five quick checks

Each should take about a minute. Answers follow.

- Two numbers are drawn independently and uniformly from  $[0, 1]$ . What is the expected distance between them?
- $X$  and  $Y$  are independent with the same variance. What is  $\text{Corr}(X, X + Y)$ ?
- A poll of 400 people has a margin of error of  $\pm 4$  points. About how many people are needed for  $\pm 2$ ?
- $X_1, \dots, X_n$  are i.i.d.  $\text{Poisson}(\lambda)$ . What is the MLE of  $\lambda$ , and is it unbiased?
- Five watches from the water test of Lecture 22 survive 2.1, 3.4, 1.7, 4.0 and 2.8 hours. Give two unbiased estimates of  $\theta$ . Which one should you report?

**Answers.**

- $\frac{1}{3}$ . The joint density is 1 on the unit square, so by symmetry  $\mathbb{E}|U_1 - U_2| = 2 \int_0^1 \int_0^x (x - y) \, dy \, dx = 2 \int_0^1 \frac{x^2}{2} \, dx = \frac{1}{3}$ . Or: the two points cut  $[0, 1]$  into three gaps of expected length  $\frac{1}{3}$  each (Lecture 13), and the distance is the middle gap.
- $\text{Cov}(X, X + Y) = \text{Var}(X) = \sigma^2$  and  $\text{Var}(X + Y) = 2\sigma^2$ , so the correlation is  $\frac{\sigma^2}{\sigma \cdot \sqrt{2}\sigma} = \frac{1}{\sqrt{2}} \approx 0.71$ .

3. About 1600. The margin of error is proportional to  $1/\sqrt{n}$ , so halving it takes four times the sample.
4.  $\ell(\lambda) = \sum_i (x_i \log \lambda - \lambda - \log x_i!)$ , so  $\ell'(\lambda) = \frac{\sum_i x_i}{\lambda} - n = 0$  gives  $\hat{\lambda} = \bar{X}$ . It is unbiased, since  $\mathbb{E}[\bar{X}] = \lambda$ .
5.  $\frac{6}{5} \max = \frac{6}{5} \cdot 4.0 = 4.8$  and  $2\bar{X} = 2 \cdot 2.8 = 5.6$ . Report 4.8: both are unbiased, but the one based on the maximum has much smaller variance (Lecture 23).

## 26.4 Worked problems

### Problem 1 (Bayes)

A disease affects 3% of the population. A test satisfies  $\mathbb{P}(+ | D) = 0.92$  and  $\mathbb{P}(- | D^c) = 0.88$ . (a) Find  $\mathbb{P}(+)$ . (b) Find  $\mathbb{P}(D | +)$ .

*Solution.*  $\mathbb{P}(D) = 0.03$ ,  $\mathbb{P}(D^c) = 0.97$ , and  $\mathbb{P}(+ | D^c) = 1 - 0.88 = 0.12$ .

$$\mathbb{P}(+) = \mathbb{P}(+ | D)\mathbb{P}(D) + \mathbb{P}(+ | D^c)\mathbb{P}(D^c) = (0.92)(0.03) + (0.12)(0.97) = 0.0276 + 0.1164 = 0.144.$$

$$\mathbb{P}(D | +) = \frac{\mathbb{P}(+ | D)\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.0276}{0.144} = 0.1917.$$

Again: a positive result raises the probability from 3% only to 19%. □

### Problem 2 (transformation)

$X \sim \text{Normal}(0, 1)$  and  $Y = e^X$ . Find  $\mathbb{P}\{Y > e^{1.5}\}$ .

*Solution.* exp is increasing, so the event is unchanged by taking logs:

$$\mathbb{P}\{Y > e^{1.5}\} = \mathbb{P}\{e^X > e^{1.5}\} = \mathbb{P}\{X > 1.5\} = 1 - \Phi(1.5) = 0.0668.$$

No density calculation is needed — that is the point of working with the cdf. □

### Problem 3 (Weibull)

$X \sim \text{Weibull}(\alpha = 3, \beta = 10)$ , so  $F(x) = 1 - e^{-(x/10)^3}$  for  $x > 0$ . (a) Find the median  $t_{0.5}$ . (b) Find  $\mathbb{P}\{5 \leq X \leq 15\}$ .

*Solution.* (a) Solve  $F(t_{0.5}) = 0.5$ :

$$1 - e^{-(t_{0.5}/10)^3} = 0.5 \Rightarrow e^{-(t_{0.5}/10)^3} = \frac{1}{2} \Rightarrow \left(\frac{t_{0.5}}{10}\right)^3 = \ln 2 \Rightarrow t_{0.5} = 10(\ln 2)^{1/3} \approx 8.85.$$

(b)  $\mathbb{P}\{5 \leq X \leq 15\} = F(15) - F(5) = e^{-(5/10)^3} - e^{-(15/10)^3} = e^{-1/8} - e^{-27/8} \approx 0.8825 - 0.0342 = 0.8483$ . □

**Problem 4 (beta)**

$X \sim \text{Beta}(3, 3)$ , so  $f(x) = 30x^2(1-x)^2$  on  $(0, 1)$ . (a) Find  $\mathbb{E}[X]$  and  $\text{Var}(X)$ . (b) Find  $\mathbb{P}\{0.4 \leq X \leq 0.6\}$ .

*Solution.* (a) By the formulas for  $\text{Beta}(\alpha, \beta)$ ,

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta} = \frac{3}{6} = \frac{1}{2}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{9}{36 \cdot 7} = \frac{1}{28}.$$

$$(b) \int_{0.4}^{0.6} 30x^2(1-x)^2 dx = 30 \left[ \frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5} \right]_{0.4}^{0.6} = 0.6826 - 0.3174 = 0.3651.$$

$\text{Beta}(3, 3)$  is also the distribution of the median of five independent uniforms (Lecture 16), which is one way to see why it is symmetric about  $\frac{1}{2}$ .  $\square$

**Problem 5 (joint pdf)**

$f_{X,Y}(x, y) = c(x + y)$  for  $0 < x < 1$ ,  $0 < y < 1$ . (a) Find  $c$ . (b) Find  $f_X$ .

*Solution.* (a)

$$1 = \int_0^1 \int_0^1 c(x + y) dx dy = c \int_0^1 \left( \frac{1}{2} + y \right) dy = c \left( \frac{1}{2} + \frac{1}{2} \right) = c \implies c = 1.$$

$$(b) f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2} \text{ for } 0 < x < 1.$$

The density is symmetric in  $x$  and  $y$ , so the same computation gives  $f_Y(y) = y + \frac{1}{2}$  on  $(0, 1)$ . Note  $f_X(x)f_Y(y) = (x + \frac{1}{2})(y + \frac{1}{2}) \neq x + y$ , so  $X$  and  $Y$  are dependent.  $\square$

**Problem 6 (covariance and correlation from a table)**

$(X, Y)$  is uniform on the six pairs  $(0, 1), (1, 1), (0, 2), (2, 2), (1, 3), (2, 3)$ , each with probability  $\frac{1}{6}$ . Find  $\mathbb{E}[X]$ ,  $\mathbb{E}[Y]$ ,  $\text{Cov}(X, Y)$ ,  $\rho_{X,Y}$ .

*Solution.* Marginals: each of  $X = 0, 1, 2$  has probability  $\frac{2}{6} = \frac{1}{3}$ , and likewise each of  $Y = 1, 2, 3$ . So

$$\mathbb{E}[X] = \frac{1}{3}(0 + 1 + 2) = 1, \quad \mathbb{E}[Y] = \frac{1}{3}(1 + 2 + 3) = 2.$$

$$\mathbb{E}[XY] = \frac{1}{6}(0 \cdot 1 + 1 \cdot 1 + 0 \cdot 2 + 2 \cdot 2 + 1 \cdot 3 + 2 \cdot 3) = \frac{14}{6} = \frac{7}{3},$$

$$\text{Cov}(X, Y) = \frac{7}{3} - (1)(2) = \frac{1}{3}.$$

$$\mathbb{E}[X^2] = \frac{1}{3}(0 + 1 + 4) = \frac{5}{3} \implies \text{Var}(X) = \frac{5}{3} - 1 = \frac{2}{3}, \quad \mathbb{E}[Y^2] = \frac{1}{3}(1 + 4 + 9) = \frac{14}{3} \implies \text{Var}(Y) = \frac{14}{3} - 4 = \frac{2}{3}.$$

$$\rho_{X,Y} = \frac{1/3}{\sqrt{(2/3)(2/3)}} = \frac{1/3}{2/3} = \frac{1}{2}.$$

$\square$

**Problem 7 (MLE)**

$X_1, \dots, X_n$  i.i.d. Exponential( $\lambda$ ),  $f(x; \lambda) = \lambda e^{-\lambda x}$ . (a) Write the likelihood. (b) Find  $\hat{\lambda}_{\text{MLE}}$ . (c) Is it unbiased?

*Solution.* (a)  $L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^n x_i\right)$ .

(b)  $\ell(\lambda) = n \log \lambda - \lambda \sum_i x_i$ , so  $\ell'(\lambda) = \frac{n}{\lambda} - \sum_i x_i = 0$  gives

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_i X_i} = \frac{1}{\bar{X}}.$$

(c) No. We know  $\mathbb{E}[\bar{X}] = 1/\lambda$ , but  $\mathbb{E}[1/\bar{X}] \neq 1/\mathbb{E}[\bar{X}]$  — the reciprocal is a non-linear function. Concretely,  $\sum_i X_i \sim \text{Gamma}(n, 1/\lambda)$  by gamma additivity, so for  $n \geq 2$

$$\mathbb{E}\left[\frac{1}{\sum_i X_i}\right] = \int_0^\infty \frac{1}{s} \cdot \frac{\lambda^n s^{n-1} e^{-\lambda s}}{\Gamma(n)} ds = \frac{\lambda \Gamma(n-1)}{\Gamma(n)} = \frac{\lambda}{n-1},$$

and therefore  $\mathbb{E}[\hat{\lambda}] = n \cdot \frac{\lambda}{n-1} = \frac{n}{n-1} \lambda > \lambda$ : the MLE overshoots on average. It is, however, asymptotically unbiased, and  $\frac{n-1}{n} \hat{\lambda}$  is exactly unbiased.  $\square$

**Remark 26.2** (Final advice). Three habits that recover the most points:

1. **Draw the region** before setting up any double integral.
2. **Go through the cdf** for anything involving a transformation.
3. **Sanity-check the answer:** probabilities lie in  $[0, 1]$ , variances are non-negative,  $|\rho| \leq 1$ , and a mean should land inside the support.